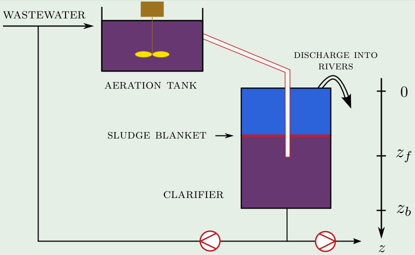
Boundary feedback control of a 2×2 weakly hyperbolic system: Lyapunov-based approach

Presented by: Mohand Ouidir Amirat

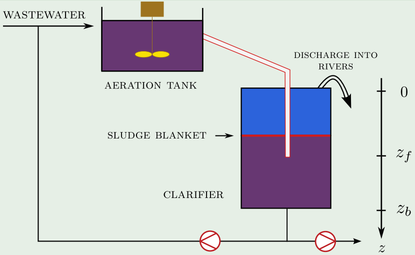
Under Supervision of:
Claire Valentin
Vincent Andrieu
Mathieu Bajodek

In Collaboration with:
Jean Auriol (L2S)

Framework


WASTEWATER
AERATION TANK
DISCHARGE INTO RIVERS
SLUDGE BLANKET
CLARIFIER
0
z_f
z_b
z

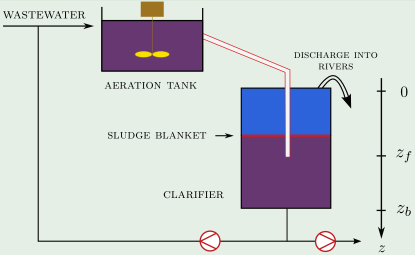
Framework



1D-Dynamic Model of the Clarifier (Two-Phase flow)

State variables: ε_s (particles density), $f_s := \varepsilon_s v_s$ (momentum) with v_s the particles velocity

Framework



1D-Dynamic Model of the Clarifier (Two-Phase flow)

State variables: ε_s (particles density), $f_s := \varepsilon_s v_s$ (momentum) with v_s the particles velocity

$$\begin{cases} \partial_t \varepsilon_s + \partial_z f_s = \dfrac{f_{1s}}{\rho_s}\delta_{z_f} \\ \partial_t f_s + \partial_z\left(\dfrac{f_s^2}{\varepsilon_s} + \dfrac{\sigma_e(\varepsilon_s)}{\rho_s}\right) = \underbrace{g\left(1 - \dfrac{\rho_l}{\rho_s}\right)\varepsilon_s}_{\text{Apparent weight}} + \underbrace{\dfrac{r(\varepsilon_s)(\varepsilon_s v_m - f_s)}{\rho_s \varepsilon_s (1-\varepsilon_s)}}_{\text{Friction}} + \underbrace{\dfrac{f_{2s}}{\rho_s}\delta_{z_f}}_{\text{Feed}} \end{cases}$$

Two boundary conditions (**BCs**) at both ends of the spatial domain are needed to close the equations. These conditions are imposed by physical devices.

$$f_s(0,t) = \mathrm{U}_0(t), \qquad f_s(z_b,t) = \mathrm{U}_1(t)$$

$\mathrm{U}_0, \mathrm{U}_1$ are *signal commands*.

Two boundary conditions (**BCs**) at both ends of the spatial domain are needed to close the equations. These conditions are imposed by physical devices.

$$f_s(0,t) = \mathrm{U}_0(t), \qquad f_s(z_b,t) = \mathrm{U}_1(t)$$

$\mathrm{U}_0, \mathrm{U}_1$ are *signal commands*.

$\longrightarrow$ Coupling the equations with (**BCs**) corresponds to a *Boundary Control System*

Two boundary conditions (**BCs**) at both ends of the spatial domain are needed to close the equations. These conditions are imposed by physical devices.

$$f_s(0,t) = \mathrm{U}_0(t), \qquad f_s(z_b,t) = \mathrm{U}_1(t)$$

$\mathrm{U}_0, \mathrm{U}_1$ are *signal commands*.

$\longrightarrow$ Coupling the equations with (**BCs**) corresponds to a *Boundary Control System*

Goal 1: Design a robust boundary controller that takes into account the Clarifier/Aeration Tank configuration (with disturbance inputs) in order to stabilize the sludge blanket depth

Goal 2: Prove the stability of the controlled system

Outline of Talk

System analysis

System analysis

Governing equations

$\mathcal{D} = [0, z_b] \times \mathbb{R}_+$ / Find ε_s, f_s: $\mathcal{D} \longrightarrow \mathbb{R}$ such that :

$$
\begin{cases}
\partial_t \varepsilon_s + \partial_z f_s = \dfrac{f_{1s}}{\rho_s}\delta_{z_f}, \\
\partial_t f_s + \partial_z\Big(\dfrac{f_s^2}{\varepsilon_s} + \dfrac{\sigma_e(\varepsilon_s)}{\rho_s}\Big) = g\Big(1 - \dfrac{\rho_l}{\rho_s}\Big)\varepsilon_s + \dfrac{r(\varepsilon_s)(\varepsilon_s v_m - f_s)}{\rho_s \varepsilon_s (1 - \varepsilon_s)} + \dfrac{f_{2s}}{\rho_s}\delta_{z_f}, \\
f_s(0, t) = \mathrm{U}_0(t), \quad f_s(z_b, t) = \mathrm{U}_1(t), \\
\varepsilon_s(z, 0) = \varepsilon_s^0, \quad f_s(z, 0) = f_s^0.
\end{cases} \tag{1}
$$

with:

$$
\sigma_e(\varepsilon_s) = \begin{cases}
0\,, & \varepsilon_s \leq \varepsilon_c \\
\sigma_0 \dfrac{\varepsilon_s^{n_s} - \varepsilon_c^{n_s}}{\varepsilon_c^{n_s}}\,, & \varepsilon_s > \varepsilon_c
\end{cases}
$$

System analysis

Governing equations

$\mathcal{D} = [0, z_b] \times \mathbb{R}_+$ / Find ε_s, f_s: $\mathcal{D} \longrightarrow \mathbb{R}$ such that :

$$\begin{cases} \partial_t \varepsilon_s + \partial_z f_s = \dfrac{f_{1s}}{\rho_s} \delta_{z_f}, \\ \partial_t f_s + \partial_z \left(\dfrac{f_s^2}{\varepsilon_s} + \dfrac{\sigma_e(\varepsilon_s)}{\rho_s}\right) = g\left(1 - \dfrac{\rho_l}{\rho_s}\right)\varepsilon_s + \dfrac{r(\varepsilon_s)(\varepsilon_s v_m - f_s)}{\rho_s \varepsilon_s (1-\varepsilon_s)} + \dfrac{f_{2s}}{\rho_s}\delta_{z_f}, \\ f_s(0,t) = \mathrm{U}_0(t), \quad f_s(z_b, t) = \mathrm{U}_1(t), \\ \varepsilon_s(z,0) = \varepsilon_s^0, \quad f_s(z,0) = f_s^0. \end{cases} \tag{1}$$

with:

$$\sigma_e(\varepsilon_s) = \begin{cases} 0\ , & \varepsilon_s \leq \varepsilon_c \\ \sigma_0 \dfrac{\varepsilon_s^{n_s} - \varepsilon_c^{n_s}}{\varepsilon_c^{n_s}}\ , & \varepsilon_s > \varepsilon_c \end{cases}$$

ρ_l: water density ($:= 1000\ \frac{kg}{m^3}$)
g: gravity acceleration constant ($:= 9.81\ \frac{m}{s^2}$)

System analysis

Governing equations

$\mathcal{D} = [0, z_b] \times \mathbb{R}_+$ / Find ε_s, f_s: $\mathcal{D} \longrightarrow \mathbb{R}$ such that :

$$\begin{cases} \partial_t \varepsilon_s + \partial_z f_s = \dfrac{f_{1s}}{\rho_s} \delta_{z_f}, \\ \partial_t f_s + \partial_z (\dfrac{f_s^2}{\varepsilon_s} + \dfrac{\sigma_e(\varepsilon_s)}{\rho_s}) = g(1 - \dfrac{\rho_l}{\rho_s})\varepsilon_s + \dfrac{r(\varepsilon_s)(\varepsilon_s v_m - f_s)}{\rho_s \varepsilon_s (1 - \varepsilon_s)} + \dfrac{f_{2s}}{\rho_s}\delta_{z_f}, \\ f_s(0,t) = \mathrm{U}_0(t), \quad f_s(z_b, t) = \mathrm{U}_1(t), \\ \varepsilon_s(z, 0) = \varepsilon_s^0, \quad f_s(z,0) = f_s^0. \end{cases} \tag{1}$$

with:

$$\sigma_e(\varepsilon_s) = \begin{cases} 0 \,, & \varepsilon_s \leq \varepsilon_c \\ \sigma_0 \dfrac{\varepsilon_s^{n_s} - \varepsilon_c^{n_s}}{\varepsilon_c^{n_s}} \,, & \varepsilon_s > \varepsilon_c \end{cases}$$

ρ_l: water density ($:= 1000 \ \frac{kg}{m^3}$)
g: gravity acceleration constant ($:= 9.81 \ \frac{m}{s^2}$)

σ_0, n_s, n_r, ε_c , τ, ρ_s (parameters)

System analysis

$$\partial_t \underbrace{\begin{pmatrix} \varepsilon_s \\ f_s \end{pmatrix}}_{\mathbf{Y}} + \partial_z \underbrace{\begin{pmatrix} f_s \\ \frac{f_s^2}{\varepsilon_s} + \frac{\sigma_e(\varepsilon_s)}{\rho_s} \end{pmatrix}}_{\mathbf{F(Y)}} = \underbrace{\begin{pmatrix} 0 \\ g(1-\frac{\rho_l}{\rho_s})\varepsilon_s + \frac{r(\varepsilon_s)(\varepsilon_s v_m - f_s)}{\rho_s\varepsilon_s(1-\varepsilon_s)} + \frac{f_{2s}}{\rho_s}\delta_{z_f} \end{pmatrix}}_{\mathbf{S(Y)}}$$

System analysis

$$\partial_t \underbrace{\begin{pmatrix} \varepsilon_s \\ f_s \end{pmatrix}}_{\mathbf{Y}} + \partial_z \underbrace{\begin{pmatrix} f_s \\ \frac{f_s^2}{\varepsilon_s} + \frac{\sigma_e(\varepsilon_s)}{\rho_s} \end{pmatrix}}_{\mathbf{F(Y)}} = \underbrace{\begin{pmatrix} 0 \\ g(1 - \frac{\rho_l}{\rho_s})\varepsilon_s + \frac{r(\varepsilon_s)(\varepsilon_s v_m - f_s)}{\rho_s \varepsilon_s (1-\varepsilon_s)} + \frac{f_{2s}}{\rho_s}\delta_{z_f} \end{pmatrix}}_{\mathbf{S(Y)}}$$

Quasi-linear form :

$$\partial_t \mathbf{Y} + \mathbf{A(Y)} \cdot \partial_z \mathbf{Y} = \mathbf{S(Y)}$$

where

$$\mathbf{A(Y)} := \frac{\partial \mathbf{F}}{\partial \mathbf{Y}} \in \mathcal{M}_2(\mathbb{R})$$

System analysis

$$\partial_t \underbrace{\begin{pmatrix} \varepsilon_s \\ f_s \end{pmatrix}}_{\mathbf{Y}} + \partial_z \underbrace{\begin{pmatrix} f_s \\ \frac{f_s^2}{\varepsilon_s} + \frac{\sigma_e(\varepsilon_s)}{\rho_s} \end{pmatrix}}_{\mathbf{F(Y)}} = \underbrace{\begin{pmatrix} 0 \\ g(1 - \frac{\rho_l}{\rho_s})\varepsilon_s + \frac{r(\varepsilon_s)(\varepsilon_s v_m - f_s)}{\rho_s \varepsilon_s (1-\varepsilon_s)} + \frac{f_{2s}}{\rho_s}\delta_{z_f} \end{pmatrix}}_{\mathbf{S(Y)}}$$

Quasi-linear form :

$$\partial_t \mathbf{Y} + \mathbf{A(Y)} \cdot \partial_z \mathbf{Y} = \mathbf{S(Y)}$$

where

$$\mathbf{A(Y)} := \frac{\partial \mathbf{F}}{\partial \mathbf{Y}} \in \mathcal{M}_2(\mathbb{R})$$

Eigenvalues of $\mathbf{A(Y)}$?

$$\mathrm{Sp}(\mathbf{A}) = \{v_s - \sqrt{\frac{\sigma_0 n_s}{\rho_s \varepsilon_c^{n_s}} \varepsilon_s^{n_s - 1}}\ ,\ v_s + \sqrt{\frac{\sigma_0 n_s}{\rho_s \varepsilon_c^{n_s}} \varepsilon_s^{n_s - 1}}\}, \ \text{ if } \varepsilon_s > \varepsilon_c$$

System analysis

$$\partial_t \underbrace{\begin{pmatrix} \varepsilon_s \\ f_s \end{pmatrix}}_{\mathbf{Y}} + \partial_z \underbrace{\begin{pmatrix} f_s \\ \frac{f_s^2}{\varepsilon_s} + \frac{\sigma_e(\varepsilon_s)}{\rho_s} \end{pmatrix}}_{\mathbf{F(Y)}} = \underbrace{\begin{pmatrix} 0 \\ g(1-\frac{\rho_l}{\rho_s})\varepsilon_s + \frac{r(\varepsilon_s)(\varepsilon_s v_m - f_s)}{\rho_s \varepsilon_s (1-\varepsilon_s)} + \frac{f_{2s}}{\rho_s}\delta_{z_f} \end{pmatrix}}_{\mathbf{S(Y)}}$$

Quasi-linear form :

$$\partial_t \mathbf{Y} + \mathbf{A(Y)} \cdot \partial_z \mathbf{Y} = \mathbf{S(Y)}$$

where

$$\mathbf{A(Y)} := \frac{\partial \mathbf{F}}{\partial \mathbf{Y}} \in \mathcal{M}_2(\mathbb{R})$$

Eigenvalues of $\mathbf{A(Y)}$?

$$\mathrm{Sp}(\mathbf{A}) = \{v_s - \sqrt{\frac{\sigma_0 n_s}{\rho_s \varepsilon_c^{n_s}} \varepsilon_s^{n_s - 1}}\,,\ v_s + \sqrt{\frac{\sigma_0 n_s}{\rho_s \varepsilon_c^{n_s}} \varepsilon_s^{n_s - 1}}\},\quad \text{if } \varepsilon_s > \varepsilon_c$$

$\Rightarrow$ Strictly hyperbolic system

System analysis

$$\partial_t \underbrace{\begin{pmatrix} \varepsilon_s \\ f_s \end{pmatrix}}_{\mathbf{Y}} + \partial_z \underbrace{\begin{pmatrix} f_s \\ \frac{f_s^2}{\varepsilon_s} + \frac{\sigma_e(\varepsilon_s)}{\rho_s} \end{pmatrix}}_{\mathbf{F(Y)}} = \underbrace{\begin{pmatrix} 0 \\ g(1-\frac{\rho_l}{\rho_s})\varepsilon_s + \frac{r(\varepsilon_s)(\varepsilon_s v_m - f_s)}{\rho_s\varepsilon_s(1-\varepsilon_s)} + \frac{f_{2s}}{\rho_s}\delta_{z_f} \end{pmatrix}}_{\mathbf{S(Y)}}$$

Quasi-linear form :

$$\partial_t \mathbf{Y} + \mathbf{A(Y)} \cdot \partial_z \mathbf{Y} = \mathbf{S(Y)}$$

where

$$\mathbf{A(Y)} := \frac{\partial \mathbf{F}}{\partial \mathbf{Y}} \in \mathcal{M}_2(\mathbb{R})$$

Eigenvalues of $\mathbf{A(Y)}$?

$$\mathrm{Sp}(\mathbf{A}) = \{v_s - \sqrt{\frac{\sigma_0 n_s}{\rho_s \varepsilon_c^{n_s}}\varepsilon_s^{n_s-1}}\ ,\ v_s + \sqrt{\frac{\sigma_0 n_s}{\rho_s \varepsilon_c^{n_s}}\varepsilon_s^{n_s-1}}\},\ \text{ if } \varepsilon_s > \varepsilon_c$$

$\Rightarrow$ Strictly hyperbolic system

$$\mathrm{Sp}(\mathbf{A}) = \{v_s\ ,\ v_s\},\ \text{ if } \varepsilon_s \leq \varepsilon_c$$

System analysis

$$\partial_t \underbrace{\begin{pmatrix} \varepsilon_s \\ f_s \end{pmatrix}}_{\mathbf{Y}} + \partial_z \underbrace{\begin{pmatrix} f_s \\ \frac{f_s^2}{\varepsilon_s} + \frac{\sigma_e(\varepsilon_s)}{\rho_s} \end{pmatrix}}_{\mathbf{F(Y)}} = \underbrace{\begin{pmatrix} 0 \\ g(1 - \frac{\rho_l}{\rho_s})\varepsilon_s + \frac{r(\varepsilon_s)(\varepsilon_s v_m - f_s)}{\rho_s \varepsilon_s (1-\varepsilon_s)} + \frac{f_{2s}}{\rho_s}\delta_{z_f} \end{pmatrix}}_{\mathbf{S(Y)}}$$

Quasi-linear form :

$$\partial_t \mathbf{Y} + \mathbf{A(Y)} \cdot \partial_z \mathbf{Y} = \mathbf{S(Y)}$$

where

$$\mathbf{A(Y)} := \frac{\partial \mathbf{F}}{\partial \mathbf{Y}} \in \mathcal{M}_2(\mathbb{R})$$

Eigenvalues of $\mathbf{A(Y)}$?

$$\mathrm{Sp}(\mathbf{A}) = \{v_s - \sqrt{\frac{\sigma_0 n_s}{\rho_s \varepsilon_c^{n_s}} \varepsilon_s^{n_s - 1}}\ ,\ v_s + \sqrt{\frac{\sigma_0 n_s}{\rho_s \varepsilon_c^{n_s}} \varepsilon_s^{n_s - 1}}\}, \quad \text{if } \varepsilon_s > \varepsilon_c$$

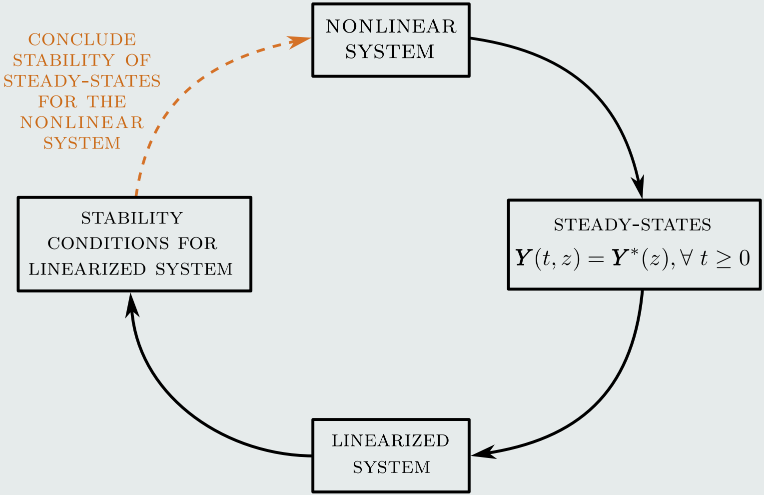
$\Rightarrow$ Strictly hyperbolic system

$$\mathrm{Sp}(\mathbf{A}) = \{v_s\ ,\ v_s\}, \quad \text{if } \varepsilon_s \leq \varepsilon_c$$

$\Rightarrow$ Weakly hyperbolic system

System analysis

How to stabilize ?



Nonuniform Shock steady-state

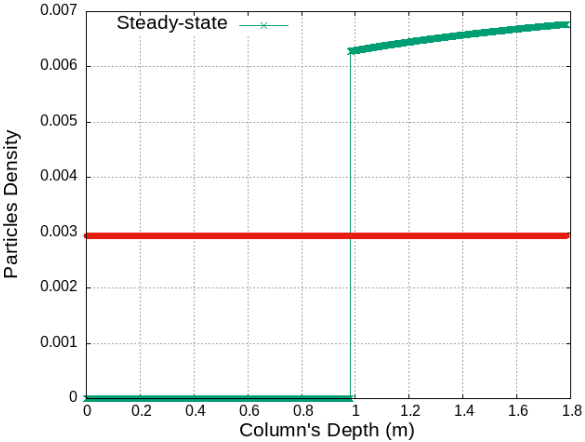


Fig. Simulation of the particles density on 6400 cells using Godunov scheme with an LLF Riemann Solver for flux computation: comparison with steady-state

System analysis: Linearization

Consider small perturbations of the states with respect to steady-state:

$$E_s := \varepsilon_s(t,x) - \varepsilon_s^*(x), \qquad V_s := v_s(t,x) - v_s^*(x)$$

System analysis: Linearization

Consider small perturbations of the states with respect to steady-state:

$$E_s := \varepsilon_s(t,x) - \varepsilon_s^*(x), \qquad V_s := v_s(t,x) - v_s^*(x)$$

Linearized system is written as:

$$\partial_t \tilde{\mathbf{Y}} + \mathbf{A}^*(z) \cdot \partial_z \tilde{\mathbf{Y}} + \mathbf{B}^*(z) \cdot \tilde{\mathbf{Y}} = 0 \tag{2}$$

$$\tilde{\mathbf{Y}} = \begin{pmatrix} E_s \\ V_s \end{pmatrix}, \quad \mathbf{A}^*(z) = \begin{pmatrix} v_s^* & \varepsilon_s^* \\ \frac{\sigma_e'(\varepsilon_s^*)}{\rho_s \varepsilon_s^*} & v_s^* \end{pmatrix}, \quad \mathbf{B}^*(z) = \begin{pmatrix} \text{Remaining} \\ \text{source terms} \end{pmatrix}$$

System analysis: Linearization

Consider small perturbations of the states with respect to steady-state:

$$E_s := \varepsilon_s(t,x) - \varepsilon_s^*(x), \qquad V_s := v_s(t,x) - v_s^*(x)$$

Linearized system is written as:

$$\partial_t \tilde{\mathbf{Y}} + \mathbf{A}^*(z) \cdot \partial_z \tilde{\mathbf{Y}} + \mathbf{B}^*(z) \cdot \tilde{\mathbf{Y}} = 0 \tag{2}$$

$$\tilde{\mathbf{Y}} = \begin{pmatrix} E_s \\ V_s \end{pmatrix}, \quad \mathbf{A}^*(z) = \begin{pmatrix} v_s^* & \varepsilon_s^* \\ \frac{\sigma_e'(\varepsilon_s^*)}{\rho_s \varepsilon_s^*} & v_s^* \end{pmatrix}, \quad \mathbf{B}^*(z) = \begin{pmatrix} \text{Remaining} \\ \text{source terms} \end{pmatrix}$$

How to decouple system (2) ?

System analysis: Linearization

Consider small perturbations of the states with respect to steady-state:

$$E_s := \varepsilon_s(t,x) - \varepsilon_s^*(x), \qquad V_s := v_s(t,x) - v_s^*(x)$$

Linearized system is written as:

$$\partial_t \tilde{\mathbf{Y}} + \mathbf{A}^*(z) \cdot \partial_z \tilde{\mathbf{Y}} + \mathbf{B}^*(z) \cdot \tilde{\mathbf{Y}} = 0 \tag{2}$$

$$\tilde{\mathbf{Y}} = \begin{pmatrix} E_s \\ V_s \end{pmatrix}, \quad \mathbf{A}^*(z) = \begin{pmatrix} v_s^* & \varepsilon_s^* \\ \frac{\sigma_e'(\varepsilon_s^*)}{\rho_s \varepsilon_s^*} & v_s^* \end{pmatrix}, \quad \mathbf{B}^*(z) = \begin{pmatrix} \text{Remaining} \\ \text{source terms} \end{pmatrix}$$

How to decouple system (2) ?

Determine a transformation that diagonalizes the matrix $\mathbf{A}^*(z)$?

System analysis: Linearization

Consider small perturbations of the states with respect to steady-state:

$$E_s := \varepsilon_s(t,x) - \varepsilon_s^*(x), \qquad V_s := v_s(t,x) - v_s^*(x)$$

Linearized system is written as:

$$\partial_t \tilde{\mathbf{Y}} + \mathbf{A}^*(z) \cdot \partial_z \tilde{\mathbf{Y}} + \mathbf{B}^*(z) \cdot \tilde{\mathbf{Y}} = 0 \tag{2}$$

$$\tilde{\mathbf{Y}} = \begin{pmatrix} E_s \\ V_s \end{pmatrix}, \quad \mathbf{A}^*(z) = \begin{pmatrix} v_s^* & \varepsilon_s^* \\ \frac{\sigma_e'(\varepsilon_s^*)}{\rho_s \varepsilon_s^*} & v_s^* \end{pmatrix}, \quad \mathbf{B}^*(z) = \begin{pmatrix} \text{Remaining} \\ \text{source terms} \end{pmatrix}$$

How to decouple system (2) ?

Determine a transformation that diagonalizes the matrix $\mathbf{A}^*(z)$?

Not uniformly possible (due to regions of weak-hyperbolicity)

System analysis: Linearization

Consider small perturbations of the states with respect to steady-state:

$$E_s := \varepsilon_s(t,x) - \varepsilon_s^*(x), \qquad V_s := v_s(t,x) - v_s^*(x)$$

Linearized system is written as:

$$\partial_t \tilde{\mathbf{Y}} + \mathbf{A}^*(z) \cdot \partial_z \tilde{\mathbf{Y}} + \mathbf{B}^*(z) \cdot \tilde{\mathbf{Y}} = 0 \tag{2}$$

$$\tilde{\mathbf{Y}} = \begin{pmatrix} E_s \\ V_s \end{pmatrix}, \quad \mathbf{A}^*(z) = \begin{pmatrix} v_s^* & \varepsilon_s^* \\ \frac{\sigma_e'(\varepsilon_s^*)}{\rho_s \varepsilon_s^*} & v_s^* \end{pmatrix}, \quad \mathbf{B}^*(z) = \begin{pmatrix} \text{Remaining} \\ \text{source terms} \end{pmatrix}$$

How to decouple system (2) ?

Determine a transformation that diagonalizes the matrix $\mathbf{A}^*(z)$?

Not uniformly possible (due to regions of weak-hyperbolicity)

Rather transform $\mathbf{A}^*(z)$ into a *Jordan* block:

$$\mathbf{A}^*(z) \longleftrightarrow \begin{pmatrix} \mathbf{c}_1 & \mathbf{c}_2 \\ \mathbf{0} & \mathbf{c}_3 \end{pmatrix}$$

Through a simplified clarifier

Target System

$$\begin{cases} \partial_t \boldsymbol{\phi}(t,x) - A(x)\partial_x \boldsymbol{\phi}(t,x) = B(x)\boldsymbol{\phi}(t,x), \; x \in (0,1), t \geq 0 \\ \phi(0,x) = \phi^0(x) \end{cases} \tag{3}$$

where $\boldsymbol{\phi} = \left(\phi_1, \phi_2\right)^\top : [0,+\infty) \times [0,1] \to \mathbb{R}^2$ is the state.

Target System

$$\begin{cases} \partial_t \boldsymbol{\phi}(t,x) - A(x)\partial_x \boldsymbol{\phi}(t,x) = B(x)\boldsymbol{\phi}(t,x),\ x \in (0,1),\ t \geq 0 \\ \phi(0,x) = \phi^0(x) \end{cases} \tag{3}$$

where $\boldsymbol{\phi} = \left(\phi_1, \phi_2\right)^\top : [0,+\infty) \times [0,1] \to \mathbb{R}^2$ is the state.
The maps $A, B : [0,1] \to \mathcal{M}_{2\times 2}(\mathbb{R})$ are assumed $C^1([0,1])$ and are defined as:

$$A(x) = \begin{pmatrix} \lambda_1(x) & c(x) \\ 0 & \lambda_2(x) \end{pmatrix}, \quad B(x) = \begin{pmatrix} B_{11}(x) & B_{12}(x) \\ B_{21}(x) & B_{22}(x) \end{pmatrix}$$

Target System

$$\begin{cases} \partial_t \boldsymbol{\phi}(t,x) - A(x)\partial_x \boldsymbol{\phi}(t,x) = B(x)\boldsymbol{\phi}(t,x), \ x \in (0,1), t \geq 0 \\ \phi(0,x) = \phi^0(x) \end{cases} \tag{3}$$

where $\boldsymbol{\phi} = \left(\phi_1, \phi_2\right)^\top : [0,+\infty) \times [0,1] \to \mathbb{R}^2$ is the state.
The maps $A, B : [0,1] \to \mathcal{M}_{2\times 2}(\mathbb{R})$ are assumed $C^1([0,1])$ and are defined as:

$$A(x) = \begin{pmatrix} \lambda_1(x) & c(x) \\ 0 & \lambda_2(x) \end{pmatrix}, \quad B(x) = \begin{pmatrix} B_{11}(x) & B_{12}(x) \\ B_{21}(x) & B_{22}(x) \end{pmatrix}$$

$c(x) \neq 0, \ \forall x \in [0,1]$, and $\lambda_{1,2}(x) > 0$

Target System

$$\begin{cases} \partial_t \boldsymbol{\phi}(t,x) - A(x)\partial_x \boldsymbol{\phi}(t,x) = B(x)\boldsymbol{\phi}(t,x), \; x \in (0,1), t \geq 0 \\ \phi(0,x) = \phi^0(x) \end{cases} \tag{3}$$

where $\boldsymbol{\phi} = \left(\phi_1, \phi_2\right)^\top : [0,+\infty) \times [0,1] \to \mathbb{R}^2$ is the state.
The maps $A, B : [0,1] \to \mathcal{M}_{2\times 2}(\mathbb{R})$ are assumed $C^1([0,1])$ and are defined as:

$$A(x) = \begin{pmatrix} \lambda_1(x) & c(x) \\ 0 & \lambda_2(x) \end{pmatrix}, \quad B(x) = \begin{pmatrix} B_{11}(x) & B_{12}(x) \\ B_{21}(x) & B_{22}(x) \end{pmatrix}$$

$c(x) \neq 0$, $\forall x \in [0,1]$, and $\lambda_{1,2}(x) > 0$ s.t. $\exists x \in [0,1]$ for which $\lambda_1(x) = \lambda_2(x)$.

Target System

$$\begin{cases} \partial_t \phi(t,x) - A(x)\partial_x \phi(t,x) = B(x)\phi(t,x), \ x \in (0,1), t \geq 0 \\ \phi(0,x) = \phi^0(x) \end{cases} \tag{3}$$

where $\phi = (\phi_1, \phi_2)^\top : [0,+\infty) \times [0,1] \to \mathbb{R}^2$ is the state.
The maps $A, B : [0,1] \to \mathcal{M}_{2\times 2}(\mathbb{R})$ are assumed $C^1([0,1])$ and are defined as:

$$A(x) = \begin{pmatrix} \lambda_1(x) & c(x) \\ 0 & \lambda_2(x) \end{pmatrix}, \quad B(x) = \begin{pmatrix} B_{11}(x) & B_{12}(x) \\ B_{21}(x) & B_{22}(x) \end{pmatrix}$$

$c(x) \neq 0, \ \forall x \in [0,1]$, and $\lambda_{1,2}(x) > 0$ s.t. $\exists x \in [0,1]$ for which $\lambda_1(x) = \lambda_2(x)$.

Previous work

Assuming $B_{21}(x) = 0$, and static boundary conditions specified as:

$$\phi(t,1) = \begin{bmatrix} D_{11} & D_{12} \\ \mathbf{0} & D_{22} \end{bmatrix} \phi(t,0) \tag{4}$$

M.O. Amirat, V. Andrieu, J. Auriol, M. Bajodek and C. Valentin, *Stability analyis of a class of 2 x 2 triangular hyperbolic systems*, 4th IFAC international conference of MICNON 2024.

How to relax coefficients $\mathbf{D}_{21}$ and $\mathbf{B}_{21}(x)$?

How to relax coefficients $\mathbf{D}_{21}$ and $\mathbf{B}_{21}(x)$?

$\Rightarrow$ Design a wider class of controllers

How to relax coefficients $\mathbf{D}_{21}$ and $\mathbf{B}_{21}(x)$?

$\Rightarrow$ Design a wider class of controllers (of *Dynamical type*)

e.g.

$$\frac{d}{dt}\begin{bmatrix}\phi_1(t,1)\\ \phi_2(t,1)\end{bmatrix} = D\begin{bmatrix}\phi_1(t,0)\\ \phi_2(t,0)\end{bmatrix} + E\begin{bmatrix}\phi_1(t,1)\\ \phi_2(t,1)\end{bmatrix} + u(t)$$

where D and E are **full** matrices, and u is the *control*.

How to relax coefficients $\mathbf{D}_{21}$ and $\mathbf{B}_{21}(x)$?

$\Rightarrow$ Design a wider class of controllers (of *Dynamical type*)

e.g.

$$\frac{d}{dt}\begin{bmatrix}\phi_1(t,1)\\ \phi_2(t,1)\end{bmatrix} = D\begin{bmatrix}\phi_1(t,0)\\ \phi_2(t,0)\end{bmatrix} + E\begin{bmatrix}\phi_1(t,1)\\ \phi_2(t,1)\end{bmatrix} + u(t)$$

where D and E are **full** matrices, and u is the *control*.

Boundary feedback control problem

$$\begin{cases}\partial_t\phi(t,x) - A(x)\partial_x\phi(t,x) = B(x)\phi(t,x),\ x\in(0,1),\\ \dot{\phi}(t,1) = D\phi(t,0) + E\phi(t,1) + u(t),\ t\in\mathbb{R}_+,\\ \phi(0,x) = \phi^0(x),\ x\in(0,1),\end{cases} \tag{5}$$

How to relax coefficients $\mathbf{D}_{21}$ and $\mathbf{B}_{21}(x)$?

$\Rightarrow$ Design a wider class of controllers (of *Dynamical type*)

e.g.

$$\frac{d}{dt}\begin{bmatrix}\phi_1(t,1)\\ \phi_2(t,1)\end{bmatrix} = D\begin{bmatrix}\phi_1(t,0)\\ \phi_2(t,0)\end{bmatrix} + E\begin{bmatrix}\phi_1(t,1)\\ \phi_2(t,1)\end{bmatrix} + u(t)$$

where D and E are **full** matrices, and u is the *control*.

Boundary feedback control problem

$$\begin{cases}\partial_t\phi(t,x) - A(x)\partial_x\phi(t,x) = B(x)\phi(t,x),\ x\in(0,1),\\ \dot{\phi}(t,1) = D\phi(t,0) + E\phi(t,1) + u(t),\ t\in\mathbb{R}_+,\\ \phi(0,x) = \phi^0(x),\ x\in(0,1),\end{cases} \tag{5}$$

Sufficient conditions for the exponential stabilizability of system (5) **when employing suitable control law**, u ?

Stability analysis of the closed-loop system

State space

$X = L^2((0,1);\mathbb{R}) \times H^1((0,1);\mathbb{R}) \times \mathbb{R}^2$ equipped with the norm:

Stability analysis of the closed-loop system

State space

$X = L^2((0,1);\mathbb{R}) \times H^1((0,1);\mathbb{R}) \times \mathbb{R}^2$ equipped with the norm:

$$\begin{aligned} \|\cdot\|_X^2 : (f,g,z) \quad &\mapsto \quad \|f\|^2_{L^2((0,1),\mathbb{R})} + \|g\|^2_{L^2((0,1);\mathbb{R})} + \|g'\|^2_{L^2((0,1);\mathbb{R})} \\ &+ \qquad z^\top z \end{aligned}$$

Stability analysis of the closed-loop system

State space

$X = L^2((0,1);\mathbb{R}) \times H^1((0,1);\mathbb{R}) \times \mathbb{R}^2$ equipped with the norm:

$$\begin{aligned} \|\cdot\|_X^2 : (f, g, z) \quad &\mapsto \quad \|f\|^2_{L^2((0,1),\mathbb{R})} + \|g\|^2_{L^2((0,1);\mathbb{R})} + \|g'\|^2_{L^2((0,1);\mathbb{R})} \\ &+ \quad z^\top z \end{aligned}$$

Exponential stabilizability

The origin of system (5) is said to be globally δ-exponentially stabilizable in X, if there exists u in $L^\infty_{loc}(\mathbb{R}_+)$, $\nu > 0$ and a decay rate $\delta > 0$ such that for any initial condition ϕ^0 satisfying compatibility conditions, we have:

$$\|\phi(t;\phi^0,u)\|_X^2 \leq \nu e^{-\delta t}\|\phi^0\|_X^2, \ \forall t \in \mathbb{R}_+. \tag{6}$$

Results

Recall:

$$A(x) = \begin{pmatrix} \lambda_1(x) & c(x) \\ 0 & \lambda_2(x) \end{pmatrix}, \quad B(x) = \begin{pmatrix} B_{11}(x) & B_{12}(x) \\ B_{21}(x) & B_{22}(x) \end{pmatrix}, \quad D \text{ and } E \text{ full matrices}$$

Results

Recall:

$$A(x) = \begin{pmatrix} \lambda_1(x) & c(x) \\ 0 & \lambda_2(x) \end{pmatrix}, \quad B(x) = \begin{pmatrix} B_{11}(x) & B_{12}(x) \\ B_{21}(x) & B_{22}(x) \end{pmatrix}, \quad D \text{ and } E \text{ full matrices}$$

Assumption

There exists a function $\mathfrak{b} \in C^1([0,1];\mathbb{R})$ such that $B_{21}(x) = \mathfrak{b}(x)(\lambda_2(x) - \lambda_1(x))$.

Results

Recall:

$$A(x) = \begin{pmatrix} \lambda_1(x) & c(x) \\ 0 & \lambda_2(x) \end{pmatrix}, \quad B(x) = \begin{pmatrix} B_{11}(x) & B_{12}(x) \\ B_{21}(x) & B_{22}(x) \end{pmatrix}, \quad D \text{ and } E \text{ full matrices}$$

Assumption

There exists a function $\mathfrak{b} \in C^1([0,1];\mathbb{R})$ such that $B_{21}(x) = \mathfrak{b}(x)(\lambda_2(x) - \lambda_1(x))$.

Theorem

Let Assumption 1 and $\frac{\overline{\lambda_2'}}{\underline{\lambda_2}} < \frac{1}{2}$ hold.
$\forall \kappa > 0$, there exist $\delta, \varepsilon_{\mathfrak{b}}, \varepsilon_B, \varepsilon_D > 0$ such that if,

$$\begin{aligned} \|D\|_2 &< \varepsilon_D, \\ \sup_{x\in(0,1)} \|B(x)\|_2 + \sup_{x\in(0,1)} \|B'(x)\|_2 &< \varepsilon_B, \\ \overline{\mathfrak{b}} + \overline{\mathfrak{b}'} &< \varepsilon_{\mathfrak{b}}, \end{aligned}$$

then the origin of the system (3) is δ-exponentially stabilizable in X with a control law given by

$$u(t) = -(\kappa \mathbb{I}_2 + E)\phi(t,1). \tag{7}$$

Sketch of the proof

The method used to prove the Theorem relies on the construction of a *nonquadratic* strict-Lyapunov functional (i.e. satisfying $\dot{\mathcal{V}}(\phi) \leq -\delta\mathcal{V}(\phi)$)

Sketch of the proof

The method used to prove the Theorem relies on the construction of a *nonquadratic* strict-Lyapunov functional (i.e. satisfying $\dot{\mathcal{V}}(\phi) \leq -\delta\mathcal{V}(\phi)$)

$$\mathcal{V} : \left\{ \begin{array}{cl} X & \to \mathbb{R}_+ \\ (\phi, z) & \mapsto (\mathcal{V}_1 + \mathcal{V}_2)(\phi) + \alpha(\mathcal{V}_3 + \mathcal{V}_4)(\phi) + \eta\mathcal{V}_5(z) \end{array} \right. \tag{8}$$

with $z = \phi(t, 1)$ and where

$$\begin{aligned}
\mathcal{V}_1(\phi) &= \int_0^1 \frac{1+\mu x}{\lambda_1(x)}\phi_1^2 dx, \ \mathcal{V}_2(\phi) = \int_0^1 \frac{1+\mu x}{\lambda_2(x)}\phi_2^2 dx \\
\mathcal{V}_3(\phi) &= \int_0^1 \frac{1+\mu x}{\lambda_2(x)}(\partial_x\phi_2)^2 dx, \\
\mathcal{V}_4(\phi) &= 2\int_0^1 \omega(x)\phi_1\partial_x\phi_2 dx, \\
\mathcal{V}_5(z) &= z^\top z.
\end{aligned}$$

parameters $\mu, \alpha, \eta \in \mathbb{R}_+^*$ and $\omega : [0, 1] \to \mathbb{R}$ that will be selected later on.

Sketch of the proof

Lemma 1 (Positivity and norm equivalence w.r.t state-space)

Let $e(\alpha)$ be given by

$$e(\alpha) = \frac{1}{2\alpha} \min \left\{ \frac{1}{\overline{\lambda_1}}, \frac{1}{\overline{\lambda_2}}, \frac{\alpha}{\overline{\lambda_2}} \right\} \tag{9}$$

Sketch of the proof

Lemma 1 (Positivity and norm equivalence w.r.t state-space)

Let $e(\alpha)$ be given by

$$e(\alpha) = \frac{1}{2\alpha} \min\left\{ \frac{1}{\overline{\overline{\lambda_1}}}, \frac{1}{\overline{\overline{\lambda_2}}}, \frac{\alpha}{\overline{\overline{\lambda_2}}} \right\} \tag{9}$$

If $\overline{\omega} \leq e(\alpha)$

Sketch of the proof

Lemma 1 (Positivity and norm equivalence w.r.t state-space)

Let $e(\alpha)$ be given by

$$e(\alpha) = \frac{1}{2\alpha} \min \left\{ \frac{1}{\overline{\lambda}_1}, \frac{1}{\overline{\lambda}_2}, \frac{\alpha}{\overline{\lambda}_2} \right\} \tag{9}$$

If $\overline{\omega} \leq e(\alpha)$, there exist positive constants, $\nu_1(\alpha, \eta)$ and $\nu_2(\alpha, \eta)$, such that, for all (ϕ, z) in X

$$\nu_1(\alpha, \eta) \|(\phi, z)\|_X^2 \leq \mathcal{V}(\phi, z) \leq \nu_2(\alpha, \eta) \|(\phi, z)\|_X^2 \tag{10}$$

where

$$\nu_1(\alpha, \eta) = \min \left(\frac{1}{2\overline{\lambda}_1}, \frac{1}{2\overline{\lambda}_2}, \frac{\alpha}{2\overline{\lambda}_2}, \eta \right), \; \nu_2(\alpha, \eta) = \max \left(\frac{\frac{3}{2} + \mu}{\underline{\lambda}_1}, \frac{1 + \mu}{\underline{\lambda}_2}, \alpha \frac{\frac{3}{2} + \mu}{\underline{\lambda}_2}, \eta \right)$$

Sketch of the proof

Time derivative of each term of $\mathcal{V}$ along C^1 solutions of system (5) yields. First,

$$\begin{aligned}
\dot{\mathcal{V}}_2(\phi) &= \underbrace{\int_0^1 (1+\mu x)\partial_x(\phi_2^2)dx}_{\text{to integrate by parts}} \\
&+ 2\int_0^1 (1+\mu x)B_{21}(x)\phi_1(x)\phi_2(x)dx \\
&+ 2\int_0^1 (1+\mu x)B_{22}(x)\phi_2^2(x)dx
\end{aligned}$$

Sketch of the proof

Time derivative of each term of $\mathcal{V}$ along C^1 solutions of system (5) yields. First,

$$\begin{aligned} \dot{\mathcal{V}}_2(\phi) &= \underbrace{\int_0^1 (1+\mu x)\partial_x(\phi_2^2)dx}_{\text{to integrate by parts}} \\ &+ 2\int_0^1 (1+\mu x)B_{21}(x)\phi_1(x)\phi_2(x)dx \\ &+ 2\int_0^1 (1+\mu x)B_{22}(x)\phi_2^2(x)dx \end{aligned}$$

Using Young's inequalities and bounds on source terms yield

$$\begin{aligned} \dot{\mathcal{V}}_2(\phi) &\leq (1+\mu)\varepsilon_B\|\phi_1\|_{L^2}^2 \\ &+ \Big(-\mu+3(1+\mu)\varepsilon_B\Big)\|\phi_2\|_{L^2}^2 \\ &+ (1+\mu)\phi_2^2(1)-\phi_2^2(0) \end{aligned} \tag{11}$$

Sketch of the proof

Second,

$$\begin{aligned} \dot{\mathcal{V}}_1(\phi) &\leq \Big(-\mu + (1+\mu)(\overline{c}\gamma_1 + 3\varepsilon_B)\Big)\|\phi_1\|^2_{L^2} \\ &+ (1+\mu)\varepsilon_B\|\phi_2\|^2_{L^2} + (1+\mu)\frac{\overline{c}}{\gamma_1}\|\partial_x\phi_2\|^2_{L^2} \\ &+ (1+\mu)\phi_1^2(1) - \phi_1^2(0) \end{aligned} \tag{12}$$

with $\gamma_1 > 0$.

Sketch of the proof

Second,

$$\begin{aligned}\dot{\mathcal{V}}_1(\phi) &\leq \Big(-\mu + (1+\mu)(\overline{c}\gamma_1 + 3\varepsilon_B)\Big)\|\phi_1\|^2_{L^2} \\ &+ (1+\mu)\varepsilon_B\|\phi_2\|^2_{L^2} + (1+\mu)\frac{\overline{c}}{\gamma_1}\|\partial_x\phi_2\|^2_{L^2} \\ &+ (1+\mu)\phi_1^2(1) - \phi_1^2(0)\end{aligned} \tag{12}$$

with $\gamma_1 > 0$.
Third,

$$\begin{aligned}\dot{\mathcal{V}}_3(\phi) &\leq (1+\mu)(1+\overline{\frac{\lambda_2'}{\lambda_2}})\varepsilon_B(\|\phi_1\|^2_{L^2} + \|\phi_2\|^2_{L^2}) \\ &+ (1+\mu)(\partial_x\phi_2(1))^2 - (\partial_x\phi_2(0))^2 \\ &+ 2\int_0^1 (1+\mu x)B_{21}(x)\partial_x\phi_1(x)\partial_x\phi_2(x)dx \\ &+ \left(-\mu + 2(1+\mu)\left(\overline{\frac{\lambda_2'}{\lambda_2}} + \varepsilon_B(2+\overline{\frac{\lambda_2'}{\lambda_2}})\right)\right)\|\partial_x\phi_2\|^2_{L^2}\end{aligned} \tag{13}$$

Sketch of the proof

Remark

The crossed term $\partial_x \phi_1 \partial_x \phi_2$ appearing in (13) cannot be handled using a Cauchy-Schwarz or Young inequality because the state ϕ_1 belongs to $L^2((0,1);\mathbb{R})$ and not to $H^1((0,1);\mathbb{R})$

Sketch of the proof

Remark

The crossed term $\partial_x\phi_1\partial_x\phi_2$ appearing in (13) cannot be handled using a Cauchy-Schwarz or Young inequality because the state ϕ_1 belongs to $L^2((0,1);\mathbb{R})$ and not to $H^1((0,1);\mathbb{R})$

$$\begin{array}{rcl} \dot{\mathcal{V}}_4(\phi) & \leq & \zeta(\varepsilon_B)(\overline{\omega}+\overline{\omega'})\|\phi\|_X^2 \\ & + & 2\int_0^1 \omega(x)(\lambda_1(x)-\lambda_2(x))\partial_x\phi_1(x)\partial_x\phi_2(x)dx \\ & + & [B_{21}(x)\omega(x)\lambda_2(x)\phi_1^2(x)]_0^1 \\ & + & 2[\omega(x)\lambda_2(x)\phi_1(x)\partial_x\phi_2(x)]_0^1 \\ & + & [\omega(x)\lambda_1(x)B_{12}(x)\phi_2^2(x)]_0^1 \end{array} \tag{14}$$

where

$$\zeta(\varepsilon_B) = 8\overline{\lambda_2}\varepsilon_B + 4\overline{\lambda_1}\varepsilon_B + 2\overline{\lambda_2} + 5\overline{\lambda_2'}\varepsilon_B + \overline{\lambda_1'}\varepsilon_B + \overline{\lambda_1}\overline{c} + 4\overline{\lambda_2'}$$

Sketch of the proof

Remark

The crossed term $\partial_x \phi_1 \partial_x \phi_2$ appearing in (13) cannot be handled using a Cauchy-Schwarz or Young inequality because the state ϕ_1 belongs to $L^2((0,1);\mathbb{R})$ and not to $H^1((0,1);\mathbb{R})$

$$\begin{array}{rcl} \dot{\mathcal{V}}_4(\phi) & \leq & \zeta(\varepsilon_B)(\overline{\omega} + \overline{\omega'})\|\phi\|_X^2 \\ & + & 2\int_0^1 \omega(x)(\lambda_1(x) - \lambda_2(x))\partial_x\phi_1(x)\partial_x\phi_2(x)dx \\ & + & [B_{21}(x)\omega(x)\lambda_2(x)\phi_1^2(x)]_0^1 \\ & + & 2[\omega(x)\lambda_2(x)\phi_1(x)\partial_x\phi_2(x)]_0^1 \\ & + & [\omega(x)\lambda_1(x)B_{12}(x)\phi_2^2(x)]_0^1 \end{array} \tag{14}$$

where

$$\zeta(\varepsilon_B) = 8\overline{\lambda_2}\varepsilon_B + 4\overline{\lambda_1}\varepsilon_B + 2\overline{\lambda_2} + 5\overline{\lambda_2'}\varepsilon_B \\ + \overline{\lambda_1'}\varepsilon_B + \overline{\lambda_1}\overline{c} + 4\overline{\lambda_2'}$$

Last we have,

$$\dot{\mathcal{V}}_5(z) = 2\eta\phi(t,1)^\top\dot{\phi}(t,1) = y^T \begin{bmatrix} 0 & \eta D^T \\ \eta D & -2\kappa\eta \end{bmatrix} y \tag{15}$$

where $y = \left[\phi(t,0), \phi(t,1)\right]^\top$.

Sketch of the proof

Summing (11)-(12)-(13)-(14)-(15) we start by properly selecting the weight function $\omega(x)$ appearing in the functional $\mathcal{V}_4$ to ensure that the crossed term $\partial_x\phi_1\partial_x\phi_2$ *vanishes*. Therefore, ω is determined such that,

$$\int_0^1 2\alpha(\lambda_1-\lambda_2)\left[\omega(x)-(1+\mu x)\mathfrak{b}(x)\right]\partial_x\phi_1\partial_x\phi_2=0$$

Sketch of the proof

Summing (11)-(12)-(13)-(14)-(15) we start by properly selecting the weight function $\omega(x)$ appearing in the functional $\mathcal{V}_4$ to ensure that the crossed term $\partial_x\phi_1\partial_x\phi_2$ *vanishes*. Therefore, ω is determined such that,

$$\int_0^1 2\alpha(\lambda_1-\lambda_2)\left[\omega(x)-(1+\mu x)\mathfrak{b}(x)\right]\partial_x\phi_1\partial_x\phi_2 = 0$$

which yields

$$\omega(x) = (1+\mu x)\mathfrak{b}(x). \tag{16}$$

Sketch of the proof

We finally get:

$$\dot{\mathcal{V}}(\phi) \leq \underbrace{\mathcal{W}(\phi)}_{\text{Part without source term}} + \underbrace{(\varepsilon_B\tilde{\zeta} + (2+3\mu)\varepsilon_{\mathfrak{b}}\zeta(\varepsilon_B))\|\phi\|_X^2}_{\text{Part w source term}} + \underbrace{y^T M y}_{\text{Bdy terms}}, \quad (17)$$

Sketch of the proof

We finally get:

$$\dot{\mathcal{V}}(\phi) \leq \underbrace{\mathcal{W}(\phi)}_{\text{Part without source term}} + \underbrace{(\varepsilon_B\tilde{\zeta} + (2+3\mu)\varepsilon_{\mathfrak{b}}\zeta(\varepsilon_B))\|\phi\|_X^2}_{\text{Part w source term}} + \underbrace{y^T My}_{\text{Bdy terms}}, \qquad (17)$$

where $\tilde{\zeta} > 0$,

$$\begin{aligned} \mathcal{W}(\phi) &= \left(-\mu + (1+\mu)\overline{c}\gamma_1\right)\|\phi_1\|_{L^2}^2 - \mu\|\phi_2\|_{L^2}^2 \\ &+ \left((1+\mu)\frac{\overline{c}}{\gamma_1} + \alpha\left[-\mu + 2(1+\mu)\frac{\overline{\lambda_2'}}{\underline{\lambda_2}}\right]\right)\|\partial_x\phi_2\|_{L^2}^2, \end{aligned}$$

Sketch of the proof

We finally get:

$$\dot{\mathcal{V}}(\phi) \leq \underbrace{\mathcal{W}(\phi)}_{\text{Part without source term}} + \underbrace{(\varepsilon_B \tilde{\zeta} + (2+3\mu)\varepsilon_{\mathfrak{b}}\zeta(\varepsilon_B))\|\phi\|_X^2}_{\text{Part w source term}} + \underbrace{y^T M y}_{\text{Bdy terms}}, \qquad (17)$$

where $\tilde{\zeta} > 0$,

$$\begin{aligned} \mathcal{W}(\phi) &= \Big(-\mu + (1+\mu)\overline{c}\gamma_1 \Big) \|\phi_1\|_{L^2}^2 - \mu \|\phi_2\|_{L^2}^2 \\ &+ \left((1+\mu)\frac{\overline{c}}{\gamma_1} + \alpha \left[-\mu + 2(1+\mu)\frac{\overline{\lambda_2'}}{\underline{\lambda_2}} \right] \right) \|\partial_x \phi_2\|_{L^2}^2, \end{aligned}$$

and,

$$M = \begin{bmatrix} K & G \\ G^T & L \end{bmatrix} \in \mathcal{M}_{4\times 4}(\mathbb{R})$$

Sketch of the proof

Lemma 2

Let $\frac{\overline{\lambda_2'}}{\underline{\lambda_2}} < \frac{1}{2}$ holds and fix

$$\mu > \max\left\{2, \frac{1 + 2\frac{\overline{\lambda_2'}}{\underline{\lambda_2}}}{1 - 2\frac{\overline{\lambda_2'}}{\underline{\lambda_2}}}\right\} \text{ and } \alpha = (1+\mu)^2\bar{c}^2. \tag{18}$$

Then, there exists a positive scalar δ such that

$$\mathcal{W}(\phi) \leq -\delta(\|\phi_1\|_{L^2}^2 + \|\phi_2\|_{L^2}^2 + \|\partial_x\phi_2\|_{L^2}^2).$$

Sketch of the proof

Lemma 2

Let $\frac{\overline{\lambda_2'}}{\underline{\lambda_2}} < \frac{1}{2}$ holds and fix

$$\mu > \max\left\{2, \frac{1 + 2\frac{\overline{\lambda_2'}}{\underline{\lambda_2}}}{1 - 2\frac{\overline{\lambda_2'}}{\underline{\lambda_2}}}\right\} \text{ and } \alpha = (1+\mu)^2 \bar{c}^2. \tag{18}$$

Then, there exists a positive scalar δ such that

$$\mathcal{W}(\phi) \leq -\delta(\|\phi_1\|_{L^2}^2 + \|\phi_2\|_{L^2}^2 + \|\partial_x \phi_2\|_{L^2}^2).$$

Then, we want to show that the quadratic form $y^\top My$ is negative definite.

Sketch of the proof

Lemma 2

Let $\frac{\overline{\lambda_2'}}{\underline{\lambda_2}} < \frac{1}{2}$ holds and fix

$$\mu > \max\left\{2, \frac{1 + 2\frac{\overline{\lambda_2'}}{\underline{\lambda_2}}}{1 - 2\frac{\overline{\lambda_2'}}{\underline{\lambda_2}}}\right\} \text{ and } \alpha = (1+\mu)^2\bar{c}^2. \tag{18}$$

Then, there exists a positive scalar δ such that

$$\mathcal{W}(\phi) \leq -\delta(\|\phi_1\|^2_{L^2} + \|\phi_2\|^2_{L^2} + \|\partial_x\phi_2\|^2_{L^2}).$$

Then, we want to show that the quadratic form $y^\top My$ is negative definite.

Approach (Schur Complement + Bound Estimates)

$$M \text{ negative definite} \Leftrightarrow \begin{cases} K \prec 0 \\ L - G^T K^{-1} G \prec 0 \end{cases} \Leftrightarrow \begin{cases} \sigma_{\max}(K) < 0 \\ \sigma_{\max}(L - G^T K^{-1} G) < 0 \end{cases}$$

Sketch of the proof

Sketch of the proof

Lemma 3

Let us define

$$m_1(\varepsilon_B) = \frac{1}{4\alpha s_1(\varepsilon_B)}, \qquad m_2(\varepsilon_B, \bar{\mathfrak{b}}) = \frac{\lambda_2(1)}{2\sqrt{\alpha(1+\mu)}}$$

$$m_3(\varepsilon_B, \bar{\mathfrak{b}}) = \frac{1}{\sqrt{2}\left[\frac{\alpha(1+\mu)}{\lambda_2^2(1)}(\kappa + \lambda_2(1)\varepsilon_B) + \alpha\bar{\mathfrak{b}}s_2(\varepsilon_B) + \eta\right]}$$

with $\eta = \frac{1}{2\kappa}\left(2 + \mu + \alpha\bar{\mathfrak{b}}(s_2(\varepsilon_2) + s_3(\varepsilon_B)) + \alpha\frac{(1+\mu)}{\lambda_2^2(1)}(\kappa + \varepsilon_B\lambda_2(1))^2\right)$.

If $\bar{\mathfrak{b}} < m_1(\varepsilon_B)$ and $\varepsilon_D < \min(m_2(\varepsilon_B, \bar{\mathfrak{b}}), m_3(\varepsilon_B, \bar{\mathfrak{b}}))$ then $\exists\delta' > 0$ such that,

$$y^\top My < -\delta'\|\phi(t,1)\|^2_{\mathbb{R}^2}.$$

End of the Proof

Based on these two Lemmas, the time derivative of the Lyapunov candidate functional along C^1-solutions of system (5) satisfies

$$\dot{\mathcal{V}}(\phi) \leq -\tilde{\delta}\mathcal{V}(\phi). \tag{19}$$

where

$$\tilde{\delta} = \frac{\min(\delta, \delta') - (\varepsilon_B \tilde{\zeta} + (2 + 3\mu)\varepsilon_{\mathfrak{b}}\zeta(\varepsilon_B))}{\nu_2(\alpha, \eta)}$$

End of the Proof

Based on these two Lemmas, the time derivative of the Lyapunov candidate functional along C^1-solutions of system (5) satisfies

$$\dot{\mathcal{V}}(\phi) \leq -\tilde{\delta}\mathcal{V}(\phi). \tag{19}$$

where

$$\tilde{\delta} = \frac{\min(\delta, \delta') - (\varepsilon_B \tilde{\zeta} + (2+3\mu)\varepsilon_{\mathfrak{b}}\zeta(\varepsilon_B))}{\nu_2(\alpha, \eta)}$$

With Grönwall inequality the above equation (19) yields

$$\mathcal{V}(\phi(t;\phi_0)) \leq e^{-\tilde{\delta}t}\mathcal{V}(\phi^0)\ ,\ \forall \phi^0 \in X.$$

End of the Proof

Based on these two Lemmas, the time derivative of the Lyapunov candidate functional along C^1-solutions of system (5) satisfies

$$\dot{\mathcal{V}}(\phi) \leq -\tilde{\delta}\mathcal{V}(\phi). \tag{19}$$

where

$$\tilde{\delta} = \frac{\min(\delta, \delta') - (\varepsilon_B \tilde{\zeta} + (2+3\mu)\varepsilon_{\mathfrak{b}}\zeta(\varepsilon_B))}{\nu_2(\alpha, \eta)}$$

With Grönwall inequality the above equation (19) yields

$$\mathcal{V}(\phi(t;\phi_0)) \leq e^{-\tilde{\delta}t}\mathcal{V}(\phi^0)\ ,\ \forall \phi^0 \in X.$$

To conclude, using the equivalence of norm stated in Lemma 1,

$$\|\phi(t;\phi^0)\|_X^2 \leq \frac{\nu_2}{\nu_1} e^{-\tilde{\delta}t}\|\phi^0\|_X^2$$

holds, which allows us to close the proof. ∎

Numerical Results

Example

Consider system (5) with

$$\lambda_1(x) = \begin{cases} 2 & \text{if } 0 \leq x < \frac{1}{2} \\ 2 - \frac{7}{2}(x - \frac{1}{2})^2 & \text{if } \frac{1}{2} \leq x \leq 1 \end{cases}$$

$$\lambda_2(x) = \begin{cases} 2, & \text{if } 0 \leq x < \frac{1}{2} \\ 2 + \frac{1}{2}(x - \frac{1}{2})^2 & \text{if } \frac{1}{2} \leq x \leq 1 \end{cases}$$

$c(x) = x + 1$, $D = \begin{bmatrix} \frac{1}{600} & \frac{1}{600} \\ \frac{1}{600} & \frac{1}{600} \end{bmatrix}$, and $\kappa = 5$. $B_{11}(x) = B_{12}(x) = B_{22}(x) = \frac{x}{3200}$, $B_{21}(x) = \frac{\lambda_2(x) - \lambda_1(x)}{3200}$.

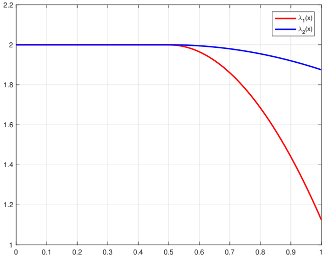
Example

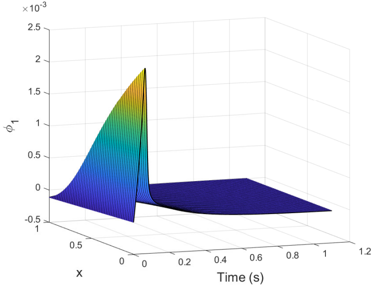
Consider system (5) with

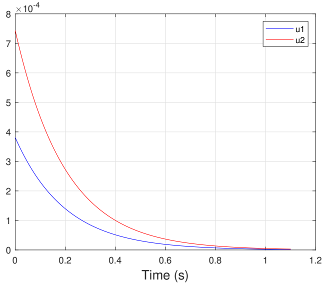
$$\lambda_1(x) = \begin{cases} 2 & \text{if } 0 \leq x < \frac{1}{2} \\ 2 - \frac{7}{2}(x - \frac{1}{2})^2 & \text{if } \frac{1}{2} \leq x \leq 1 \end{cases}$$

$$\lambda_2(x) = \begin{cases} 2, & \text{if } 0 \leq x < \frac{1}{2} \\ 2 + \frac{1}{2}(x - \frac{1}{2})^2 & \text{if } \frac{1}{2} \leq x \leq 1 \end{cases}$$

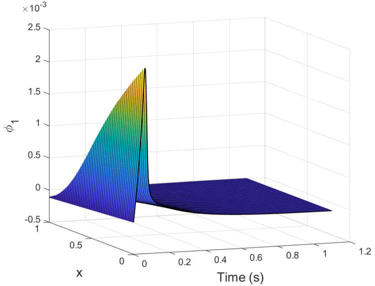
$c(x) = x + 1$, $D = \begin{bmatrix} \frac{1}{600} & \frac{1}{600} \\ \frac{1}{600} & \frac{1}{600} \end{bmatrix}$, and $\kappa = 5$. $B_{11}(x) = B_{12}(x) = B_{22}(x) = \frac{x}{3200}$, $B_{21}(x) = \frac{\lambda_2(x) - \lambda_1(x)}{3200}$.

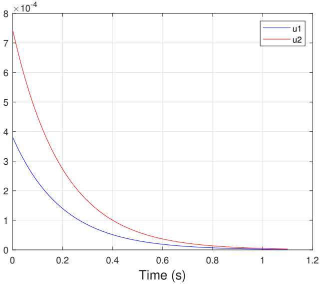




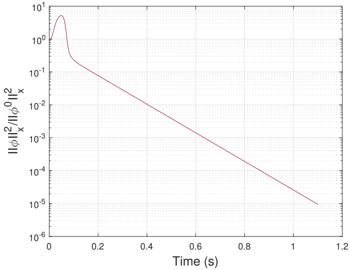


System (5): (left) space-time evolution of state ϕ_1 / (right) time evolution of boundary control $u(t)$





System (5): (left) space-time evolution of state ϕ_1 / (right) time evolution of boundary control $u(t)$



Time evolution of $\frac{\|\phi\|_X^2}{\|\phi^0\|_X^2}$ in semi-logarithmic scale

Remarks

i) Both states are stabilized to zero as time grows up.

Remarks

i) Both states are stabilized to zero as time grows up.

ii) With reference to Theorem 1, the assumptions on matrices D and B are satisfied, as their norms are indeed small

Remarks

i) Both states are stabilized to zero as time grows up.

ii) With reference to Theorem 1, the assumptions on matrices D and B are satisfied, as their norms are indeed small

iii) Furthermore, we have numerically computed the decay rate and overshoot in these simulations and obtained:

$$\delta_{\text{num}} = 10.01,$$
$$\nu_{\text{num}} = 0.58.$$

Conclusions and perspectives

Conclusions

Study of a class of Non-Strictly Hyperbolic problems:

Conclusions

Study of a class of Non-Strictly Hyperbolic problems:

$\longrightarrow$ Tackling the *new* difficulties related to the non-diagonal structure

Conclusions

Study of a class of Non-Strictly Hyperbolic problems:

$\longrightarrow$ Tackling the *new* difficulties related to the non-diagonal structure

$\longrightarrow$ Establishing sufficient stability conditions for this class of systems

Conclusions

Study of a class of Non-Strictly Hyperbolic problems:

$\longrightarrow$ Tackling the *new* difficulties related to the non-diagonal structure

$\longrightarrow$ Establishing sufficient stability conditions for this class of systems

$\longrightarrow$ Full Lyapunov analysis

Conclusions

Study of a class of Non-Strictly Hyperbolic problems:

$\longrightarrow$ Tackling the *new* difficulties related to the non-diagonal structure

$\longrightarrow$ Establishing sufficient stability conditions for this class of systems

$\longrightarrow$ Full Lyapunov analysis

$\longrightarrow$ Numerical validation and comparison with theoretical results

Conclusions

Study of a class of Non-Strictly Hyperbolic problems:

$\longrightarrow$ Tackling the *new* difficulties related to the non-diagonal structure

$\longrightarrow$ Establishing sufficient stability conditions for this class of systems

$\longrightarrow$ Full Lyapunov analysis

$\longrightarrow$ Numerical validation and comparison with theoretical results

Note that Non-Strictly Hyperbolic equations are of important interest and arise in many physical areas such as **Wastewater Treatment**,

Conclusions

Study of a class of Non-Strictly Hyperbolic problems:

$\longrightarrow$ Tackling the *new* difficulties related to the non-diagonal structure

$\longrightarrow$ Establishing sufficient stability conditions for this class of systems

$\longrightarrow$ Full Lyapunov analysis

$\longrightarrow$ Numerical validation and comparison with theoretical results

Note that Non-Strictly Hyperbolic equations are of important interest and arise in many physical areas such as **Wastewater Treatment**, Elasticity theory,

Conclusions

Study of a class of Non-Strictly Hyperbolic problems:

$\longrightarrow$ Tackling the *new* difficulties related to the non-diagonal structure

$\longrightarrow$ Establishing sufficient stability conditions for this class of systems

$\longrightarrow$ Full Lyapunov analysis

$\longrightarrow$ Numerical validation and comparison with theoretical results

Note that Non-Strictly Hyperbolic equations are of important interest and arise in many physical areas such as **Wastewater Treatment**, Elasticity theory, Advection of tracers in pipes etc.

Conclusions

Study of a class of Non-Strictly Hyperbolic problems:

$\longrightarrow$ Tackling the *new* difficulties related to the non-diagonal structure

$\longrightarrow$ Establishing sufficient stability conditions for this class of systems

$\longrightarrow$ Full Lyapunov analysis

$\longrightarrow$ Numerical validation and comparison with theoretical results

Note that Non-Strictly Hyperbolic equations are of important interest and arise in many physical areas such as **Wastewater Treatment**, Elasticity theory, Advection of tracers in pipes etc.

Paper submissions

M.O. Amirat, V. Andrieu, J. Auriol, M. Bajodek and C. Valentin, *Stability analyis of a class of 2 x 2 triangular hyperbolic systems*, 4th IFAC international conference of MICNON 2024.

M.O. Amirat, V. Andrieu, J. Auriol, M. Bajodek and C. Valentin, *Boundary feedback control of a class of 2×2 weakly hyperbolic systems*, 5th IFAC Workshop on Control of Systems Governed by PDEs (CPDE 2025).
Paper Status: Accepted

Perspectives

Ongoing work

Drawback: Use of Cauchy-Schwarz/Young inequalities brings high conservatism in estimation of decay rates, overshoots in the Lyapunov analysis.

Perspectives

Ongoing work

Drawback: Use of Cauchy-Schwarz/Young inequalities brings high conservatism in estimation of decay rates, overshoots in the Lyapunov analysis.

$\Rightarrow$ Relax the conservatism by expressing stability conditions in terms of **Matrix inequalities** (c.f. J.-M. Coron and G. Bastin), *work in preparation...*

Perspectives

Ongoing work

Drawback: Use of Cauchy-Schwarz/Young inequalities brings high conservatism in estimation of decay rates, overshoots in the Lyapunov analysis.

$\Rightarrow$ Relax the conservatism by expressing stability conditions in terms of **Matrix inequalities** (c.f. J.-M. Coron and G. Bastin), *work in preparation...*

2) Apply this theory to the Clarifier/Aeration tank and conclude stability of the nonlinear system.

Thank you for your attention
Questions ?